

DAY 03: KEY PAIRS, SECURITY GROUPS, EBS, INSTANCE STORE & FILE COPY

Cloud Computing & DevOps Master Curriculum

1. SSH Key Pairs & Authentication

AWS EC2 uses public-key cryptography to secure login credentials. A **Key Pair** consists of a **Public Key** (stored by AWS on the instance) and a **Private Key file (.pem)** (kept securely on your local computer).

Connecting to Linux EC2 via SSH:

```
chmod 400 my-key.pem
ssh -i my-key.pem ec2-user@
```

2. Security Groups (Virtual Firewalls)

A **Security Group** acts as a virtual firewall for your EC2 instance to control incoming (Inbound) and outgoing (Outbound) traffic.

- Security Groups are **Stateful**: If inbound traffic is allowed, outbound response traffic is automatically allowed.
- All Inbound traffic is blocked by default until explicit rules are added.
- Common ports: SSH (22), HTTP (80), HTTPS (443), RDP (3389), Custom App (8080).

3. Storage Comparison: EBS vs Instance Store

EBS (Elastic Block Store)	Instance Store Volume
Persistent (Survives instance stop/delete)	Ephemeral (Data lost on stop/terminate)
Network-attached block storage	Physically attached to host server
High (SSD/HDD with IOPS options)	Ultra-high IOPS / Lowest latency
Databases, Boot volumes, Core files	Temporary caches, buffers, scratch data
Supports point-in-time EBS Snapshots	No native snapshot support

4. File Transfer & Copy Commands

Copy file from Local PC to EC2 Instance (SCP):

```
scp -i my-key.pem app.zip ec2-user@:/home/ec2-user/
```

Copy file from EC2 Instance to AWS S3 (AWS CLI):

```
aws s3 cp /home/ec2-user/backup.tar.gz s3://my-cloud-bucket-name/
```

Sync local folder to remote server:

```
rsync -avz -e 'ssh -i my-key.pem' ./dist/ ec2-user@:/var/www/html/
```

KEY TAKEAWAYS & SUMMARY

- ✓ Never share private key (.pem) files; enforce 400 permissions on Linux/macOS.
- ✓ Security groups evaluate rules at the ENI (Elastic Network Interface) level statefully.
- ✓ Use EBS for persistent root/data storage; use Instance Store only for transient, ultra-fast temporary caches.